CALEB CHAI YINN LIANG

Embedded Systems | PCB Design | Motor Control | Robotics Hardware
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Malaysian | Expected Graduation: July 2026 | Portfolio: calebchai.github.io

PROFILE

Automation undergraduate at Zhejiang University with hands-on experience in embedded systems, PCB design, sensor acquisition, motor control, robotics hardware, and robot imitation-learning workflows. Experienced in schematic design, firmware development, debugging, competition deployment, and real robotic applications.

EDUCATION

Zhejiang University, College of Control Science and Engineering - B.Eng. in Automation, expected July 2026

Relevant coursework: Artificial Intelligence and Machine Learning, Aerial Robotics, Motion Control, Robot Modeling and Control, Embedded Systems, Automatic Control Theory, Signals and Systems, Analog Electronics, Electrical Control, DSP Technology, FPGA Systems, Sensing and Detection.

TECHNICAL SKILLS

Programming	C, C++, Python
Embedded	STM32, ESP32-S3, Keil 5, STM32CubeIDE, Arduino IDE, ESP-IDF, SPI, UART, I2C, CAN
PCB & Hardware	KiCad, Altium Designer, JLCEDA, schematic design, PCB layout, soldering, hardware debugging
Control & Robotics	FOC motor control, PID tuning, current-loop control, state machines, sensor processing, VESC, IMU, Raspberry Pi
Data & Tools	MATLAB, Matplotlib, PyTorch, Q-learning, Isaac Lab, RSL-RL, GVHMR, GMR

GRADUATION DESIGN PROJECT

Electric Roller Speed Control System Using Open-Source VESC and Q-Learning PID — 2025 - 2026

B.Eng. Graduation Design

- Built an electric-roller speed-control system using the open-source VESC platform, focusing on controller integration, PID speed-loop tuning, and validation rather than self-designed motor-drive hardware.
- Explore Q-learning-based PID parameter tuning for acceleration/deceleration, sudden acceleration, emergency braking, and constant-speed scenarios.
- Build MATLAB/Python simulation and debugging workflow to evaluate tracking response, overshoot, stability, and recovery under different operating conditions.

EXPERIENCE & PROJECTS

Hangzhou K2 Co. Ltd. — Jan 2025 - Present

Embedded and Circuit Intern

- Participate in shuttle-car and electric-roller controller development, covering sensor processing, state-machine design, and FOC motor-control implementation.
- Built a video-to-robot imitation-learning workflow using GVHMR for human motion recovery, GMR for motion retargeting, and RSL-RL/Isaac Lab for robot policy training and simulation validation.
- Designed STM32-based controller solutions to replace traditional industrial PLCs, reducing hardware cost while keeping the control system deployable in real equipment.
- Built control strategies for real operating conditions, including current-loop closed-loop control, status feedback, multi-sensor processing, and on-machine parameter tuning.

Hangzhou Turing Robotics Co.Ltd. — Mar 2024 - Jan 2025

Embedded and Circuit Intern

- Designed and debugged schematic/PCB boards, soldered prototypes, and developed embedded firmware for pressure-sensing gloves, six-axis force sensors, and four-wheel robots.
- Built an ESP32-S3 sensor acquisition system using a dedicated acquisition IC; implemented SPI communication in C on ESP-IDF, transmitted data via UART, and visualized data with Python Matplotlib.
- Migrated robot schematic/PCB designs from JLCEDA to KiCad, upgraded input voltage from 16 V to 24 V, and integrated the open-source VESC motor-control solution.
- Delivered a six-axis force sensor for robotic-arm end-effector perception; reduced the robot motor-control system cost to 20% of the previous design while maintaining performance.

ZJUNlict Small-Size Soccer Robot Laboratory — Sep 2023 – Sep 2025

Circuit Team Core Member

- Lead circuit-team project management, PCB design, soldering, embedded firmware, and hardware integration for small-size soccer robots.
- Redesigned motor communication and control by replacing I2C with CAN and adding current-loop feedback; increased motor-control frequency from 200 Hz to 1 kHz.
- Integrated IMU sensing and transmitted feedback to Raspberry Pi; redesigned ball-position sensing from 1D detection to a 2D reflective infrared array using I2C.
- Contributed to the ZJUNlict 2025 Extended Team Description Paper (ETDP), submitted on 4 Nov 2025; documented robot hardware architecture, control system, motor communication, sensor integration, and competition-system design.
- **Awards: 2024 RoboCup World Cup Small-Size League runner-up; RoboCup China Open national first prize; RoboCup Japan Open third place; 2024 Zhejiang Robot Competition provincial second prize.**

12th Zhejiang Provincial University Student Intelligent Car Competition — Jan 2023 - Aug 2023

Team Member

- Responsible for hardware setup, sensor selection, and debugging; adapted and optimized the provided PCB and hardware in Altium Designer for competition use.
- Selected and tuned infrared ranging sensors for obstacle detection and avoidance; the stable PCB was used in the provincial competition and won the provincial third prize.

ADDITIONAL

Languages: Mandarin Chinese native; Malay and English for basic communication. Strengths: end-to-end engineering execution across PCB design, embedded firmware, debugging, and deployed robotic systems.